

colour and polarising filters in addition to a liquid crystal layer that block, bend, and vary the intensity of light to produce images. It's the resulting gap between the backlight and the various layers that introduces colour shift and other picture quality imperfections when viewed from extreme angles. Newer LCD technologies such as IPS panels have greatly improved off-axis screen performance, but nothing comes close to the viewing angles achieved by OLEDs. Because these panels emit light – as opposed to LEDs blocking it – they offer excellent off-axis performance, with virtually no colour and gamma uniformity issues even when viewed from extremely acute angles.

Thus far we have seen the rather severe limitations posed by LED-backlit LCD panels and how OLEDs fill in these gaps with excellent picture quality and performance. However, no display technology is perfect and OLEDs too have teething troubles and minor niggles. In the next instalment we will take a look at these OLED-specific issues and, more importantly, how premium LEDs strike back with quantum dot technology that promises to deliver emissive-like performance in the existing transmissive technology.

A Matter of Resolution

Resolution has different ramifications depending upon the type of application. IPS Monitors generally ship in wider screen size variation with generous resolutions such as 2560 x 1440 instead of 1920 x 1080 for 27-inch models and 2560 x 1600 for 30-inch models. But before you make that sort of commitment, make sure that your gaming rig is good enough to

